Fiscal Year:	FY 2008	**Task Last Updated:**	FY 11/18/2008
PI Name:	Hlatky, Lynn Ph.D.		
Project Title:	Computational Modeling of Chromosome Aberrations Produced by HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02135-2907	**Congressional District:**	8
Comments:	PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: grant # changed in 2006 when PI moved--new number is NNJ06HA27G; also, end date is now 9/30/2010, per S. Krenek/JSC (3/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hahnfeldt, Philip (Caritas St. Elizabeth's Medical Center, Tufts Univ School of Medicine) Sachs, Rainer (University of California)		
Grant/Contract No.:	NNJ06HA27G ; NNJ04HF42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Chromosome aberrations produced by high LET radiations are directly and indirectly relevant to space exploration and risk estimation in a number of different ways. They are important as probes for DNA repair and recombination mechanisms, as biomarkers of dose and radiation quality, as factors in genomic instability and carcinogenesis, and as surrogates for cancer in risk estimation. Quantitative modeling is needed to exploit and augment the rapidly accumulating empirical data on aberration types and frequencies. This project will apply mathematical and computer algorithms to data on aberrations produced by iron ions (56Fe). The analysis will be compared to analyses of damage produced by photons, electrons, or alpha particles, with the goal of assessing signature responses to 56Fe, an important radiation encountered in space. To perform such studies, the team has tied damage endpoints to risk through our customized software and algorithms. These endpoints include: 1. aberration complexity, quantified by cycle structures; 2. apparently and truly incomplete aberrations, related to segment size distributions; 3. aberration spectra as related to chromosome geometry, to DNA double strand break (DSB) clustering depending on radiation track structure, and especially to DSB misrejoining depending on proximity; 4. mFISH color junctions for each specific chromosome pair, related to chromosome-chromosome spatial correlations in an interphase nucleus. Substantial preliminary studies have recently been carried out by our team for each aim. As remarked, we will employ the team's specialized and comprehensive software, including the highly-developed chromosome aberration simulator (CAS) software, for analyzing the aberration spectra encountered. The data to be analyzed will be for human cells in vitro, scored by mFISH or other similar techniques. A number of these data sets are being made available by many of our colleagues worldwide. This will greatly expand the range of suitable data available for analysis. We have also recently devised methods for systematically comparing mFISH data with data obtained by older methods such as FISH or solid staining and with newer methods such as color banding. For Specific Aim 1, we will replace the current approach of using "obligate" (deterministic) cycles with a more accurate computer-based approach involving cycle probabilities. For Aim 2, new methods are being developed that allow systematic comparison of apparently incomplete aberrations with corresponding complete aberrations and corresponding truly incomplete ones, even in the case of very complex aberrations. Under Aim 2 we will also consider size distributions of fragments that have one telomere and one unrejoined free end, for comparison to recent data using pan-telomeric probes; and analyze mixtures of different radiations. Aim 3 will emphasize more realistic computer algorithms for break misrejoining, bringing this aspect of aberration modeling more nearly in line with the very sophisticated models now being used by various groups for assessing the influence of track structure and chromatin geometry on DSB locations. For Aim 4 we plan to apply a method we previously used for low LET data to data for high LET, including iron ions. The method involves testing frequencies of color junction pairs for deviations from randomness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved biodosimetry
Task Progress:	Progress for the reporting period, 05/16/2007-05/15/2008. We further compared complexity for iron ion MFISH data with alpha particle data (kindly supplied by M. Cornforth) and analyzed incompleteness. The results extended those reported in the paper by Levy et al. (2007), reported in the bibliography. We continued to make further progress on Aim 2, on DSB (DNA double-strand break) clustering at high LET and its influence on misrejoining frequencies. We have recently collaborated with A. Ponomarev and F. Cucinotta of NASA in work on how to subtract out background values of DNA fragment size distributions in PFGE experiments on radiation breakage to analyze DSB yields and clustering. The group has found a fully general subtraction method, based on stochastic point-process theory, which works for non-random (high LET) double strand break induction as well as random (low LET) double strand break induction and for random or non-random background breakage. The method is also applicable to subtracting out low LET effects on break distributions in the case of mixed HZE radiations and protons on DSB clustering. The results were reported in a peer reviewed paper that appeared in this reporting period, Ponomarev et al. (2007). We began the mathematical analysis of aberrations which appear during cancer progression, especially in aneuploidy and in the cell-cell fusion as recently observed in our lab. Such aberrations often involve the loss, gain, or replacement (e.g. via loss of heterozygosity) of whole chromosomes or of chromosome segments so large that they are visible or visibly missing optically at metaphase. Mathematically what is required is an extension of the aberration multigraph formalism that relaxes the requirement of conservation of genetic material. We have developed a tentative formalism using a single "reservoir" from which extra chromosome segments are drawn and to which missing segments are relegated, with appropriate initial or final edges in the aberration multigraph imposed. We utilized the linear-quadratic dose-dependence formalism for radiation damage formation, suggested by kinetic studies on aberration formation as corroborated by CAS, to analyze incidence of second cancers after tumor radiotherapy, taking into account cell population dynamics for the heterogeneous cell population during and after fractionated protocols and stochastic effects implied by the fact that in early stages the number of pre-malignant cells can be very small. The results were reported in a paper in the Journal of Theoretical Biology. Cumulative bibliography with updates 1) Full-Color Painting Reveals an Excess of Radiation-Induced Dicentrics Involving Homologous Chromosomes. Y. Plan, L. Hlatky, P. Hahnfeldt, R. Sachs, B. Loucas, and M. Cornforth. Int J Radiat Biol 2005; 81:613-20. 2) Levy D, L Hlatky and RK Sachs (2006), Software for Cycle Analysis, http://radiobiology.berkeley.edu/ NSF 3) SCHIP: Statistics for Chromosome Interphase Positioning. Vives S, Loucas B, Vazquez M, Brenner DJ, Sachs RK, Hlatky L, Cornforth M, Arsuaga J. Bioinformatics. 2005 21:3181-2. 4) Ponomarev, A. L., Belli, M., Hahnfeldt, P. J., Hlatky, L., Sachs, R. K., and Cucinotta, F. A., 2006, A robust procedure

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9) Human X chromosome reveals pronounced early population subdivisions and recent bottlenecks Allen M Chen1, Jack S Chen2, Lynn Hlatky3, Caoan Wang4, Dan Levy1, Mariel Vazquez1,5, Rainer K Sachs1. Submitted to Plos1.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Ponomarev AL, Belli M, Hahnfeldt PJ, Hlatky L, Sachs RK, Cucinotta FA. "Subtraction of background damage in PFGE experiments on DNA fragment-size distributions." Radiation and Environmental Biophysics. 2007 Jun;46(2):155-60. PMID: 17406880 , Jun-2007
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Articles in Peer-reviewed Journals	Levy D, Reeder C, Loucas B, Hlatky L, Chen A, Cornforth M, Sachs R. "Interpreting chromosome aberration spectra." Journal of Computational Biology. 2007 Mar;14(2):144-55. PMID: 17456013 , Mar-2007
Articles in Peer-reviewed Journals	Chen AM, Chen JS, Hlatky L, Wang C, Levy D, Vazquez M, Sachs RK. "Human X chromosome reveals pronounced early population subdivisions and recent bottlenecks." Submitted to PLoS Biology, March 2008. , Mar-2008